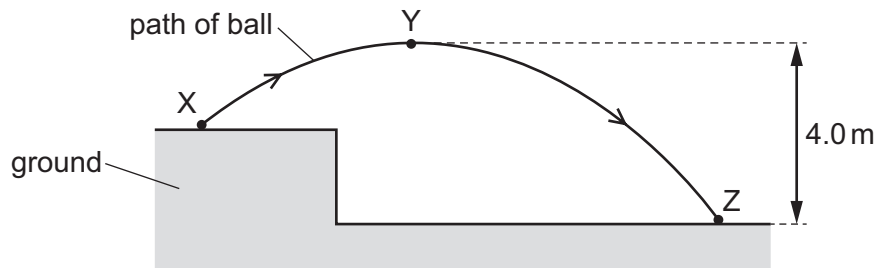


- 16** A ball is projected from the ground at point X. The ball then reaches its maximum height at point Y before landing on lower ground at point Z. The diagram shows the path of the ball.



not to scale

The ball leaves the ground at X with a kinetic energy of 5.2 J. The ball has a kinetic energy of 9.6 J when it hits the ground at Z.

The weight of the ball is 2.0 N. The vertical height of Y above Z is 4.0 m.

Air resistance is negligible.

What is the vertical height of Y above X?

- A** 1.8 m **B** 2.0 m **C** 2.2 m **D** 2.6 m